FIT5196 A1 — Group030

Focused EDA from six reconciled relational tables

7 assessed figures · exactly 10 findings · exactly 5 ML questions

1. Context and preparation assurance

JSON and XML were parsed structurally before bounded regex. Dates, timestamps, booleans, currency and percentage points were normalised before primary-key reconciliation. Order arithmetic follows rounded lines, included GST, discount, then delivery; multilingual and Latin-only text are separate (MAP-*).

Executable checks cover ordered schemas and required values (VAL-SCHEMA/MISS-*), PK/FK integrity, raw-to-canonical overlap (VAL-FLOW-*), $0.01 arithmetic, ranges, temporal order, literal NaN, reference formats and Unicode measures. EDA joins specify cardinality and assert unchanged left-table row counts.

2. Assessed-figure analytical contracts

Figure 1 — Question: what is typical versus high-value? Unit/denominator: each of 5,000 orders. Table: orders. Limitation: value is not profit.

Figure 2 — Question: do deeper coupons correspond to larger gross baskets? Unit: order within discount group. Table: orders. Means use 95% CIs. Limitation: observational coupon assignment.

Figure 3 — Question: which categories contribute estimated gross margin? Unit: 15,723 item lines. Join: items.product_id → products.product_id (many-to-one; row count retained). Limitation: catalogue cost and no line discount allocation.

Figure 4 — Question: are monthly changes volume- or value-driven? Unit: month from all orders. Table: orders. Limitation: one year and unequal month lengths.

Figure 5 — Question: does historical frequency carry forward? Unit: 500 customers. Join: customer-level orders → customers.customer_id (one-to-one). Limitation: linear unadjusted association.

Figure 6 — Question: which carrier/service cells warrant review? Unit: delivery within cell. Join: deliveries.order_id → orders.order_id (one-to-one). Wilson 95% CIs. Limitation: uncontrolled route mix.

Figure 7 — Question: is OTIF associated with rating? Unit: order with ≥1 review. Reviews are aggregated by order_id, then joined one-to-one to deliveries. Point and 95% CI use the full 1–5 scale. Limitation: reviewer selection and product confounding.

Figure 1. Order-value distribution and high-value threshold

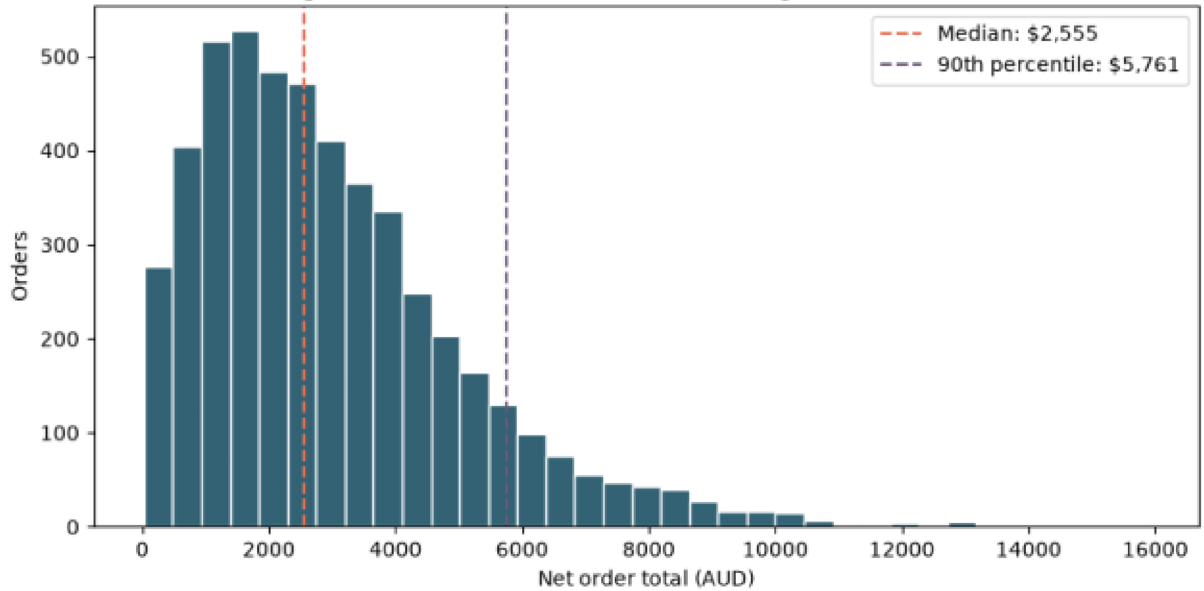
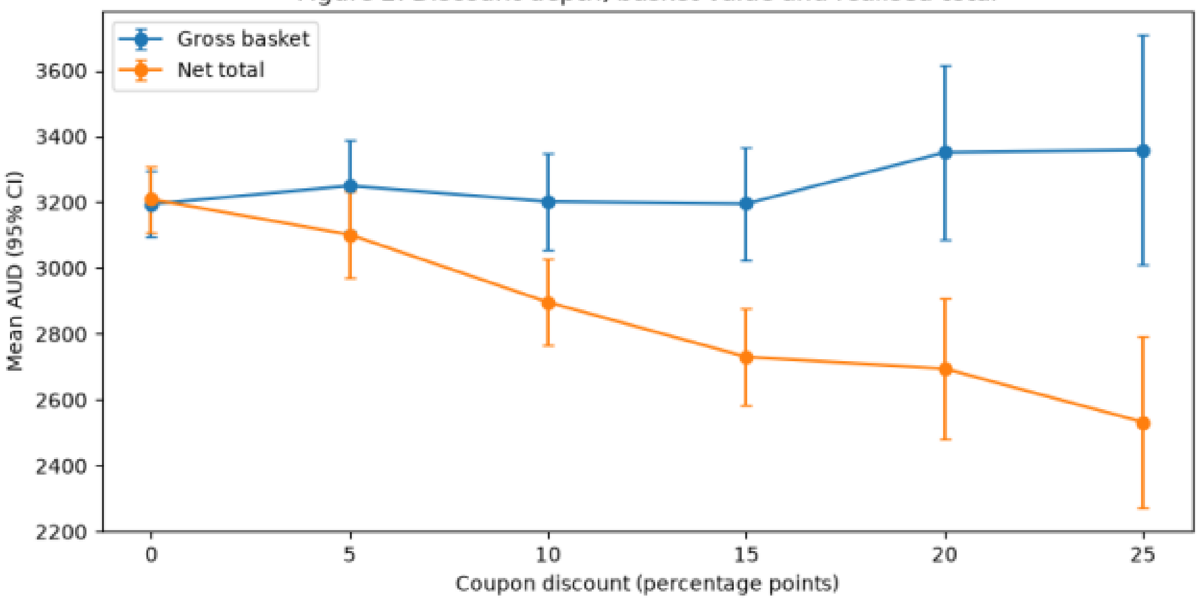


Figure 2. Discount depth, basket value and realised total



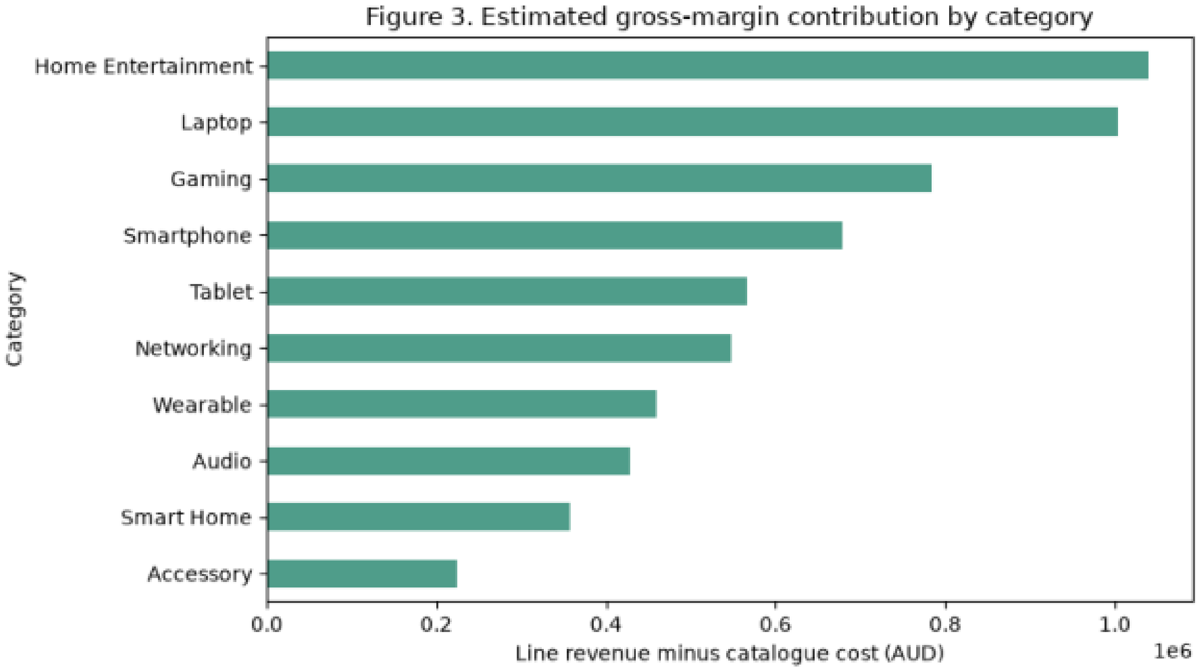


Figure 3. Estimated gross-margin contribution by category
Home Entertainment
Laptop
Gaming
Smartphone
Tablet
Networking
Wearable
Audio
Smart Home
Accessory
Category
0.0
0.2
0.4
0.6
0.8
1.0
1e6
Line revenue minus catalogue cost (AUD)

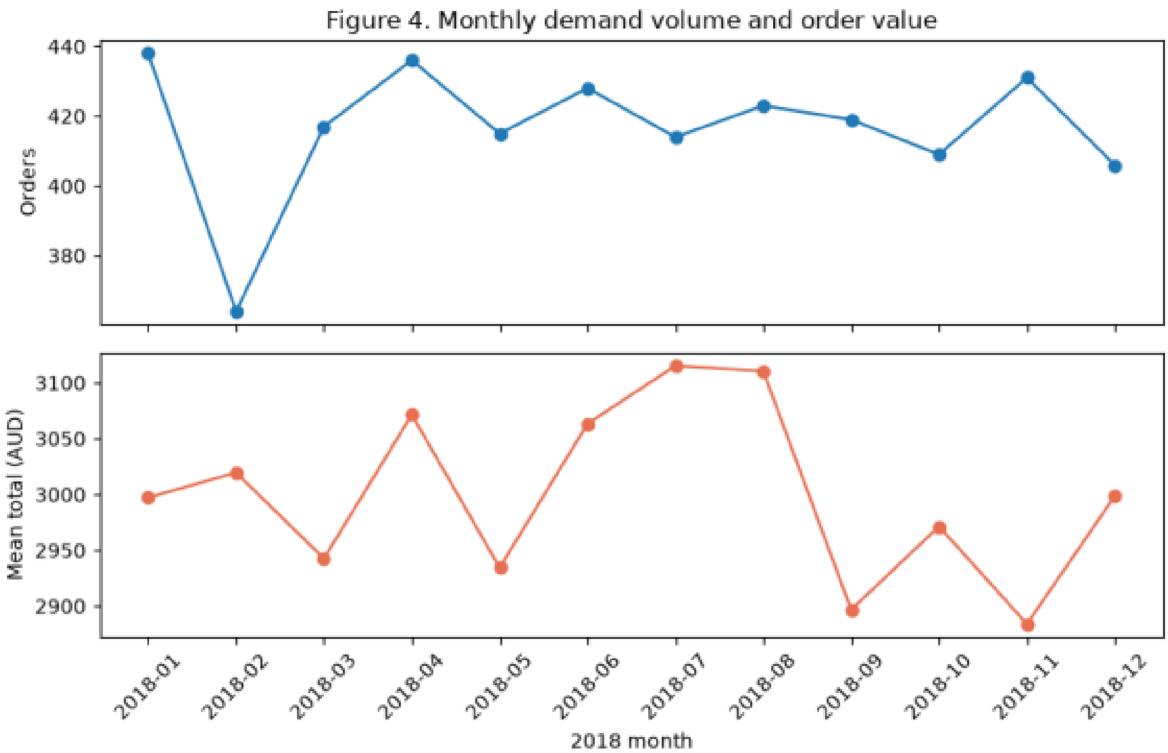


Figure 4. Monthly demand volume and order value
Orders
440
420
400
380
Mean total (AUD)
3100
3050
3000
2950
2900
2018-01
2018-02
2018-03
2018-04
2018-05
2018-06
2018-07
2018-08
2018-09
2018-10
2018-11
2018-12
2018 month

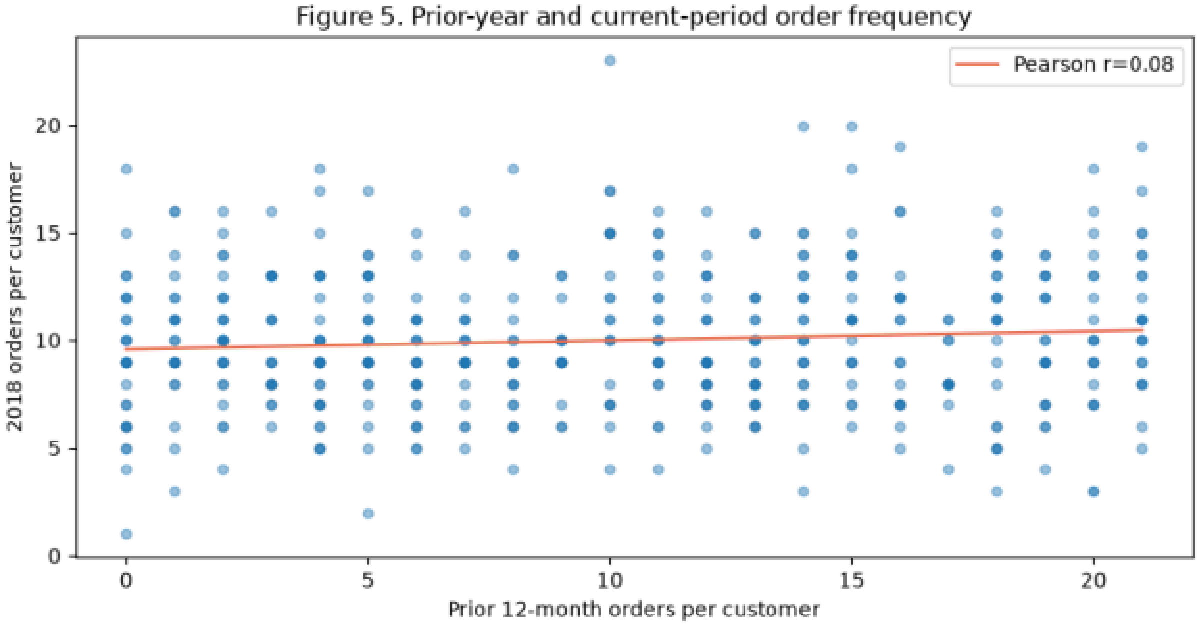


Figure 5. Prior-year and current-period order frequency
Pearson r=0.08
2018 orders per customer
Prior 12-month orders per customer
0
5
10
15
20

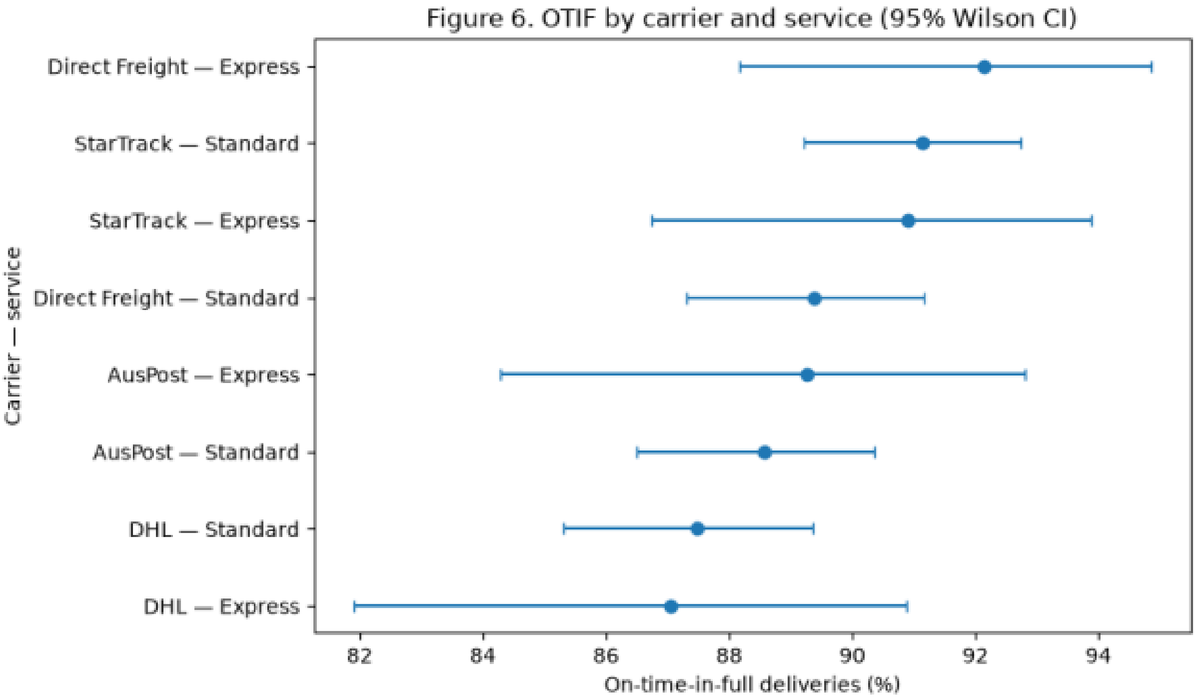


Figure 6. OTIF by carrier and service (95% Wilson CI)
Direct Freight — Express
StarTrack — Standard
StarTrack — Express
Direct Freight — Standard
AusPost — Express
AusPost — Standard
DHL — Standard
DHL — Express
Carrier — service
82
84
86
88
90
92
94
On-time-in-full deliveries (%)

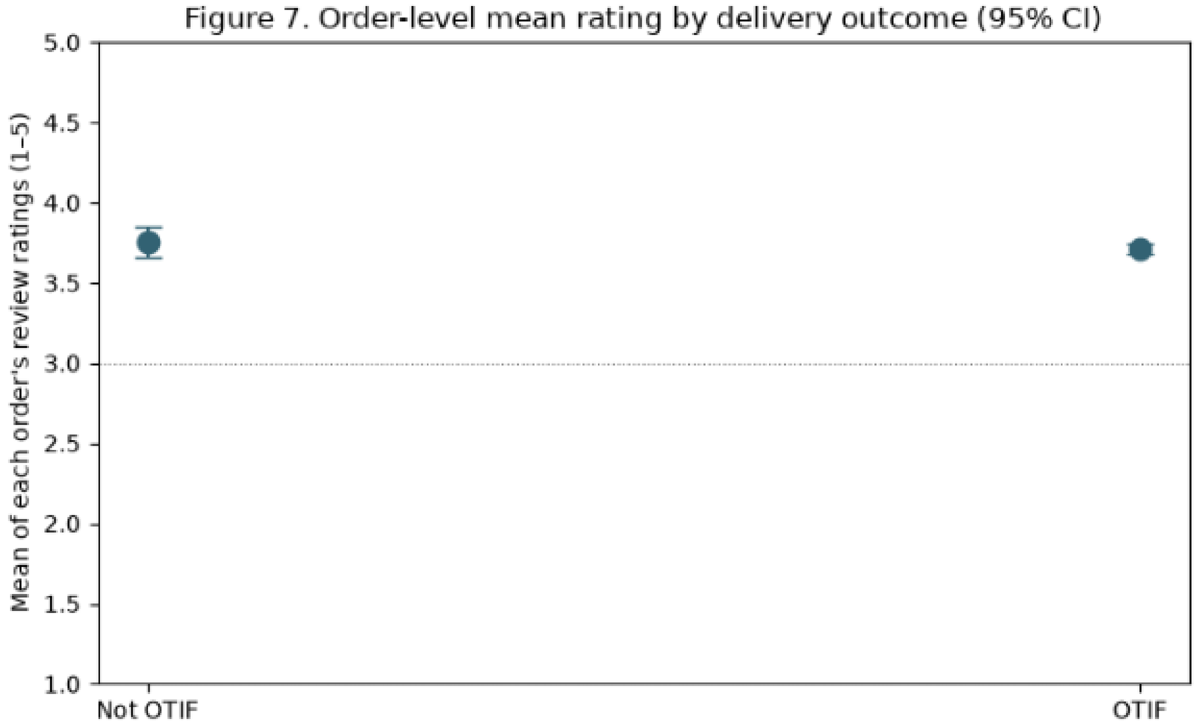


Figure 7. Order-level mean rating by delivery outcome (95% CI)
Mean of each order's review ratings (1–5)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Not OTIF
OTIF

3. Ten evidence-based findings

1. Figure 1 — Across 5,000 orders, median net total was $2,555.44; the top 10% began at $5,760.76. The long upper tail makes the mean unrepresentative of a typical order. Category mix may explain high values, so use this threshold for investigation, not automatic treatment.

2. Figure 2 — Gross basket means stayed within $3,195–$3,359, while net total fell from $3,209 at 0% to $2,533 at 25% (n=156). Coupon assignment is not random; test causal lift before increasing depth.

3. Figure 3 — Home Entertainment contributed the most estimated gross margin ($1,039,705) across 1,190 lines. Catalogue cost excludes overhead, returns and discounts, so this ranks contribution rather than accounting profit; add realised costs before assortment changes.

4. Figure 4 — Monthly volume ranged from 364 to 438 orders and monthly AOV from $2,884 to $3,115. One year cannot separate seasonality from promotions or month length; obtain several years and campaign calendars.

5. Figure 5 — For 500 customers, prior and current order counts had Pearson r=0.08, a weak positive association. History alone has limited ranking power; tenure and campaign exposure may explain changes. Validate a multivariable model temporally.

6. Figure 6 — Overall OTIF was 89.3% across 5,000 deliveries, leaving material improvement scope. Aggregate performance hides carrier/service mix; target investigation at stratum level.

7. Figure 6 — The lowest observed cell was DHL Express at 87.0% OTIF (n=216). Its interval overlaps peers, so rank is not proof of underperformance; control for route mix and collect repeated periods.

8. Figure 7 — Order-level mean rating differed by only 0.04 points between OTIF and non-OTIF outcomes across 4,044 rated orders. This near-null average does not prove delivery is unimportant: selection and product quality may mask effects. Examine low-rating risk and delay severity.

9. Figure 3 — Networking had the highest estimated margin rate (39.9%) but contributed $547,039, well below Home Entertainment, because scale differs. Rate-only ranking could over-prioritise a smaller pool; use both contribution and rate, subject to unallocated discounts.

10. Text assurance — 303 of 7,000 reviews contained non-Latin letters. Removing them would discard customer evidence. Preserve multilingual text and evaluate future text models by language, acknowledging small-group uncertainty.

4. Five future ML questions

MLQ-1 — OTIF-risk classification. Decision/unit/target: proactive intervention per order / OTIF failure. Predictors available at dispatch: warehouse, carrier, service, distance, expedited flag, basket and timing. Rolling temporal split; PR-AUC, calibration and recall at capacity. Exclude delivered date, delay reason and tracking outcomes; audit geographic/service false negatives.

MLQ-2 — Fulfilment-hours regression. Decision/unit/target: dispatch workload planning per order / fulfilment_hours. Use warehouse, order timestamp, expedited flag, service level, basket line count, quantity, value and shipping distance available by fulfilment start. Use forward temporal splits; MAE and 90th-percentile absolute error. Dispatch/delivery dates and tracking events leak outcomes; monitor error by warehouse and service level.

MLQ-3 — Low-rating classification. Decision/unit/target: service recovery per completed order / 1–2 star review. Use product, customer history and delivery facts available before outreach; temporal split, PR-AUC and calibration. Text, helpful votes and rating-derived fields leak; reviewer selection and language fairness matter.

MLQ-4 — Customer-frequency regression. Decision/unit/target: retention capacity per customer / next-90-day count. Use past history, tier, acquisition and past mix; cohort temporal split, MAE and Poisson deviance. Exclude future orders/updated lifetime value; audit protected proxies.

MLQ-5 — Product clustering. Decision/unit/objective: assortment roles per product / stable demand-value profiles using lagged volume, revenue, margin, rating and price. Test time-window stability, silhouette and merchant interpretability. Scaling and sparse exposure may dominate.

5. Limitations and conclusion

The export is observational, covers one retailer-year, and contains repeated customers/products. Coupon and delivery assignments are not random; reviews condition on reviewers; catalogue cost is not realised cost; delivery dates are day-level; subgroup sizes differ. Associations are not causal.

The clearest decision signals are category economic concentration and heterogeneous OTIF. Discount depth does not visibly expand gross baskets, but causal lift needs an experiment. Historical frequency is weak alone and average ratings barely differ by OTIF. Future work requires temporal validation, leakage control and subgroup monitoring.